Gradient Minds — Learning Rate Scheduling Research Library

Research Paper Index & Evidence Reading Report

IT3071 – Machine Learning and Optimization Methods | Assignment 1
Group 14 | Gradient Minds | Affirmative Position

Debate Statement: “Cosine Learning Rate Decay Outperforms Step Decay Scheduling.”

This report organizes the eight papers reviewed for the submitted Evidence Portfolio and records their objectives, methodologies, key evidence, debate relevance and limitations.

Important: The evidence is interpreted as a qualified research conclusion. Step Decay can outperform Cosine under particular conditions; the portfolio does not claim universal superiority.

1. Eigencurve

Full reference: Pan, R., Ye, H., & Zhang, T. (2022). “Eigencurve: Optimal Learning Rate Schedule for SGD on Quadratic Objectives with Skewed Hessian Spectrums.” ICLR 2022.

Objective: Explain theoretically why learning-rate schedules differ and why cosine can outperform step decay.

Methodology: Theoretical convergence analysis plus controlled experiments involving ResNet-18, GoogLeNet and VGG16, with CIFAR-10 and additional ImageNet experiments.

Key evidence: On CIFAR-10, Cosine exceeded Step for all three listed architectures at both 10 and 100 epochs. For ResNet-18 at epoch 10: Step 73.77% vs Cosine 84.23%. The portfolio also records ImageNet evidence favouring cosine.

Debate value: Combines direct Cosine-vs-Step evidence with a theoretical explanation. Important qualification: the paper also studies other schedules, so it does not establish universal cosine superiority.

2. A Second Look at Exponential and Cosine Step Sizes

Full reference: Li, X., Zhuang, Z., & Orabona, F. (2021). “A Second Look at Exponential and Cosine Step Sizes: Simplicity, Adaptivity, and Performance.” ICML 2021, PMLR 139.

Objective: Provide convergence guarantees for cosine/exponential schedules, investigate noise adaptivity, and compare practical performance and tuning complexity.

Methodology: Mathematical proofs; FashionMNIST, CIFAR-10, CIFAR-100; SNLI natural-language inference; synthetic noise experiments.

Key evidence: Cosine was best or tied-best on 2 of 3 image datasets and performed strongly across all three. The portfolio also highlights fewer scheduling hyperparameters and noise-adaptive behaviour.

Debate value: Strong theory + practical evidence. Qualification: some accuracy margins over carefully tuned step decay are narrow.

3. REX

Full reference: Chen, J., Wolfe, C., & Kyrillidis, A. (2022). “REX: Revisiting Budgeted Training with an Improved Schedule.” MLSys 2022.

Objective: Revisit learning-rate scheduling under fixed training budgets and compare established schedules.

Methodology: Empirical comparisons across tasks and training budgets.

Key evidence: The portfolio reports cosine achieving equal or better performance than step in multiple settings and showing more consistent behaviour across budgets, while step remains competitive in some long-budget scenarios.

Debate value: Supports flexibility and consistency of smooth scheduling. Qualification: the paper is primarily about REX, so it is supporting rather than definitive cosine-vs-step evidence.

4. Bag of Tricks

Full reference: He, T., Zhang, Z., Zhang, H., Zhang, Z., Xie, J., & Li, M. (2019). “Bag of Tricks for Image Classification with Convolutional Neural Networks.” CVPR Workshops.

Objective: Evaluate practical training improvements, including learning-rate scheduling.

Methodology: Empirical CNN experiments including ResNet-50, Inception-V3 and MobileNet in ImageNet-style classification settings.

Key evidence: Reported improvements include ResNet-50-D 77.16%→77.91%, Inception-V3 77.50%→78.19%, and MobileNet 71.90%→72.83% when cosine scheduling was used as a refinement. Transfer-learning object detection also improved.

Debate value: Practical multi-architecture evidence. Qualification: focused mainly on computer vision.

5. Large Scale Performance Analysis

Full reference: Aach, M., Inanc, E., Sarma, R., Riedel, M., & Lintermann, A. (2023). “Large Scale Performance Analysis of Distributed Deep Learning Frameworks for Convolutional Neural Networks.” Journal of Big Data, 10, 96.

Objective: Study stepwise, cosine and exponential annealing in large-scale distributed CNN training.

Methodology: ResNet50/101/152 on ImageNet, SGD, distributed HPC training and batch sizes from 256 to 65,536.

Key evidence: Cosine achieved superior validation performance to stepwise scheduling at small and medium batch sizes. At batch size 65k, exponential was best and stepwise exceeded cosine.

Debate value: Direct evidence that cosine can outperform step under common training conditions. Qualification: explicitly shows scheduler performance depends on training conditions.

6. SGDR

Full reference: Loshchilov, I., & Hutter, F. (2017). “SGDR: Stochastic Gradient Descent with Warm Restarts.” ICLR 2017.

Objective: Introduce cosine annealing with warm restarts to improve training efficiency and performance.

Methodology: CIFAR-10, CIFAR-100, downsampled ImageNet and EEG classification using Wide Residual Networks, compared with traditional scheduling.

Key evidence: The portfolio reports lower CIFAR errors than the default step schedule and comparable/better results with roughly 2–4 times fewer training epochs.

Debate value: Foundational evidence for cosine annealing. Qualification: SGDR includes warm restarts, so it should not be presented as proof that plain cosine always beats step.

7. Systematic Evaluation

Full reference: Mateen, H., Timofte, R., & Ignatov, D. (2026). “Systematic Evaluation of Learning Rate Scheduling Strategies Across Heterogeneous Architectures.” arXiv:2607.08511.

Objective: Systematically evaluate scheduler effects across heterogeneous neural-network architectures.

Methodology: 30 architectures, 25 scheduler configurations, 9 scheduler families and 3,938 evaluated variants on CIFAR-10.

Key evidence: CosineAnnealingLR mean top-1 accuracy 54.55% vs StepLR 53.70%; best reported configuration 63.97% vs 62.79%. Architecture-specific results also show Step winning for some models.

Debate value: Excellent breadth evidence against the claim that cosine wins only in one special model. Qualification: five-epoch/CIFAR-10 scope and architecture-dependent results prevent a universal claim.

8. Stepsize Anything

Full reference: Tang, A., Dong, Y., Zeng, Y., Zhou, X., & Lin, Z. (2025). “Stepsize Anything: A Unified Learning Rate Schedule for Budgeted-Iteration Training.” NeurIPS 2025.

Objective: Develop a theoretically motivated unified schedule for fixed training budgets and compare it with Step and Cosine.

Methodology: CIFAR-10/ResNet18, CIFAR-100/ResNet34 and ImageNet/ResNet50 at 25%, 50% and 100% budgets, plus optimizer and initial-learning-rate analyses.

Key evidence: Cosine exceeded Step at all budgets on CIFAR-10/ResNet18 and ImageNet/ResNet50. However, Step exceeded Cosine on CIFAR-100/ResNet34 by 6.97, 5.50 and 2.95 percentage points at 25%, 50% and 100%.

Debate value: Particularly valuable because it contains both supporting and contradictory evidence in one study. It demonstrates that dataset, architecture, budget and initial LR matter.

Overall Research Interpretation

The eight-paper portfolio combines theoretical analysis, controlled Cosine-vs-Step experiments, practical deep-learning studies, multi-dataset evaluations, budgeted-training studies and large-scale scheduler comparisons.

The strongest direct evidence comes from studies that compare Cosine and Step under the same experimental framework. The strongest breadth evidence comes from the systematic evaluation and Stepsize Anything. SGDR is best treated as foundational cosine-annealing evidence because it includes warm restarts.

The most important counter-evidence is the CIFAR-100/ResNet34 result in Stepsize Anything and architecture-specific StepLR wins in the systematic evaluation. These results were retained to support a balanced, evidence-based debate.

Recommended conclusion: The research supports Cosine as a strong and often more consistent alternative to traditional Step Decay across many evaluated settings, but does not establish universal superiority for every dataset, architecture, optimizer, budget or hyperparameter configuration.